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JDBC (PART 1)

Terms you should know before JDBC :-

1. Programming Language :- A programming language is a computer language programmers use to develop software programs, scripts, or other sets of instructions for computers to execute. Examples are C, C++, Java, Python, C#

2. Technologies :- Technology is the sum of techniques, skills, methods and processes used in the production of goods or services or in the accomplishment of objectives, such as scientific investigation. For example there are a lot of java technologies like JDBC, Servlets, JSP, EJB, JNDI, JavaMail

3. Package :- Package is the collection of similar type of classes or interfaces or packages

4. API :- (Application Programming Interface) - It contains deep documentation about software or product or technology (collection of similar type of classes, interfaces and packages) eg. Java Speech API, JSF, Java OpenGL, JavaMail API, JPA

5. DBMS :- The *database management system* (*DBMS*) is the software that interacts with end users, applications, and the database itself to capture and analyze the data.

6. SQL :- *SQL* stands for Structured Query Language. *SQL* lets you access and manipulate databases.

7. Database :- A database is an organized collection of data, generally stored and accessed electronically from a computer

system. For example MySQL, Oracle database, PostgreSQL, IBM DB2 etc.

8. JDBC :- Explained below…

8.1. What is JDBC :-

- **JDBC is an abstraction** which is provided by Sun Microsystems and is implemented by all database vendors (Driver) (Abstraction is the process of hiding certain details and showing only essential information to the user. Abstraction can be achieved with either abstract classes or interfaces)
- **JDBC is a technology** because it interacts the Java Application with database
- **JDBC is an API** (i.e. JDBC contains some classes and interfaces and packages)
  There are 2 packages :-
  1. java.sql 2. javax.sql

" java.sql " package		" javax.sql " package	
Interfaces	Classes	Interfaces	Classes
• Driver • Connection • Statement • PreparedStatement • CallableStatement • ResultSet	• DriverManager • Date • Time • TimeStamp • Types	• DataSource • RowSet • RowSetListener • ConnectionEventListener	• RowSetEvent • ConnectionEvent

8.2 History and Versions of JDBC :-

First version of JDBC was released on February, 1997 i.e version JDBC 1.0 by Sun Microsystems

Versions are as follows :-

- Java SE 1.1 : JDBC 1.0
- Java SE 1.4 : JDBC 3.0
- Java SE 5 : No updates
- Java SE 6 : JDBC 4.0
- Java SE 7 : JDBC 4.1
- Java SE 8 : JDBC 4.2

8.3 Features Of JDBC :-

- JDBC provides an API thru which we can communicate with database
- Mostly database drivers are developed in java thus these are platform independent
- Easy to use CRUD operation
- Using JDBC we can request any type of database query
- JDBC support advanced datatypes i.e BLOB, CLOB etc
- Using JDBC we can set save points for database and also we can rollback to desired save points
- Using JDBC we can send multiple updates to database and this is known as batch updating

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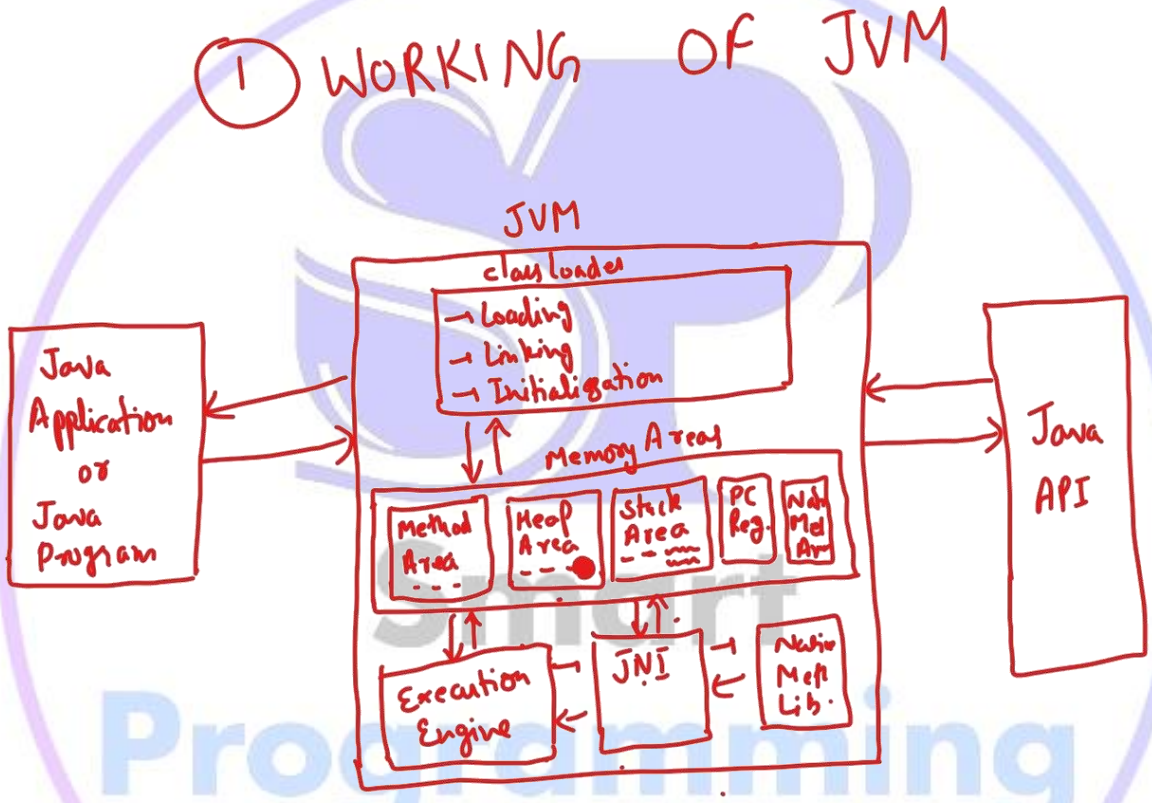
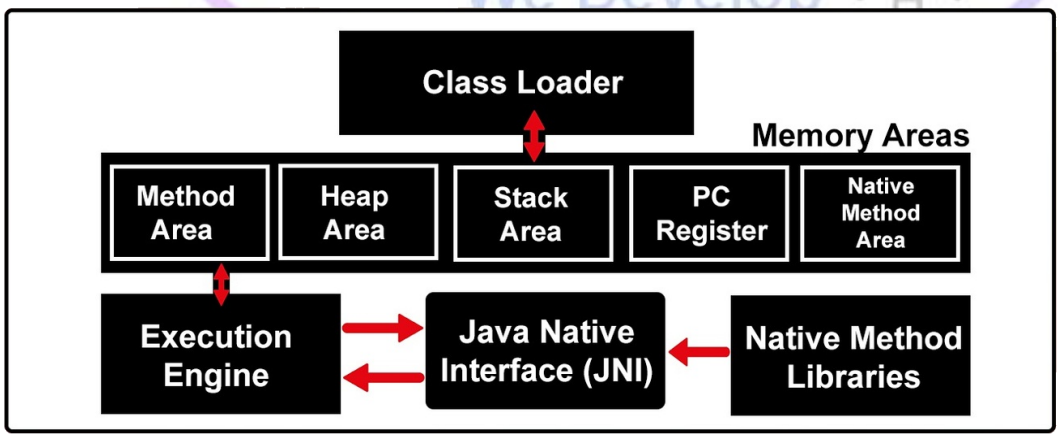
8.4 Working of JDBC :-

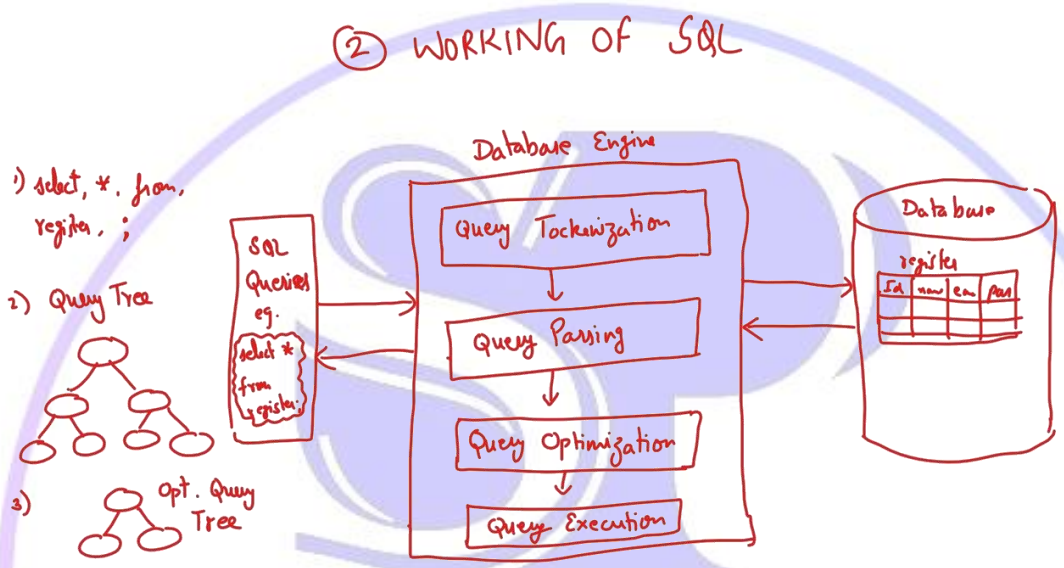
- **Working of JVM**



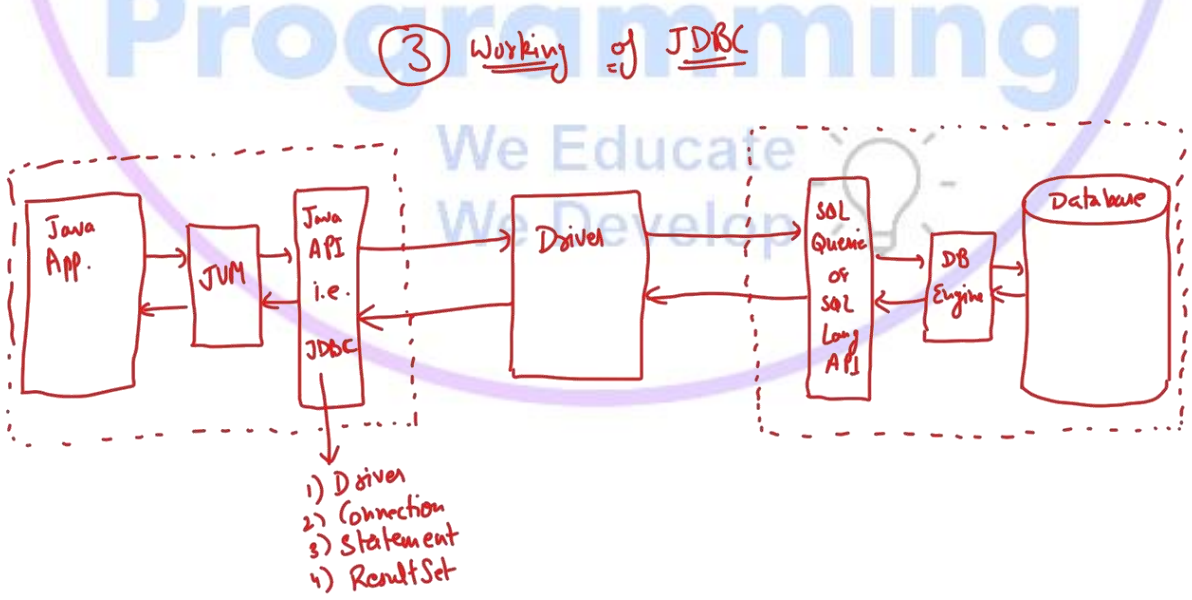
JVM Architecture



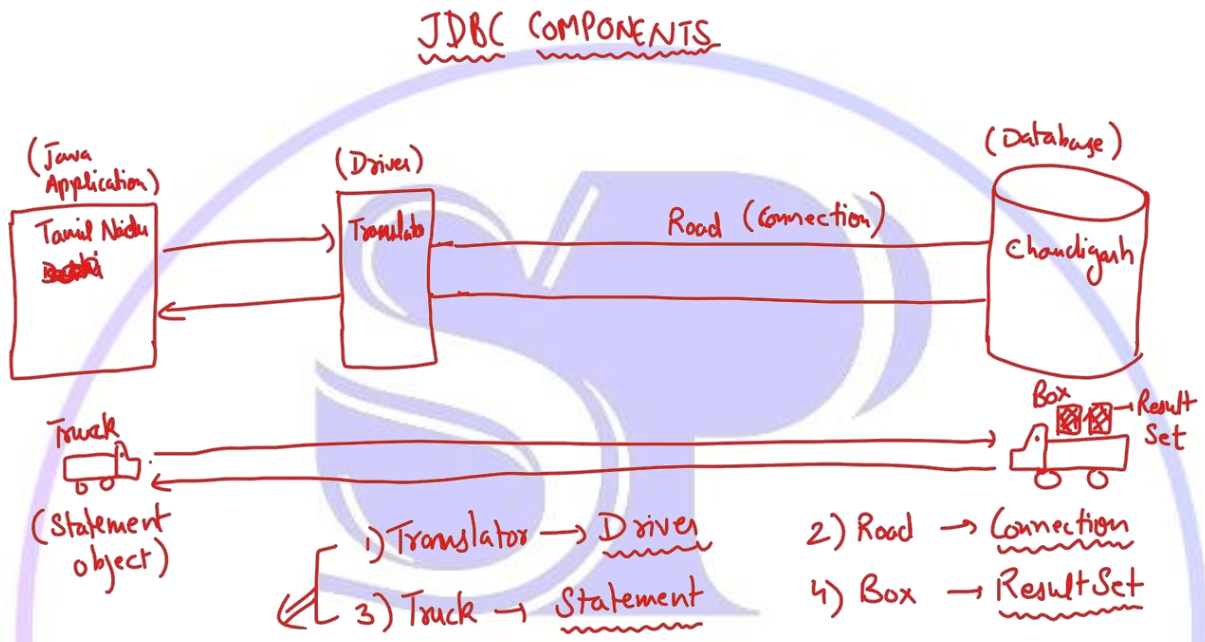
- **Working of SQL**



- **Working of JDBC**



- **Explanation of Working of JDBC through real world**



8.5 JDBC Components :

- **Driver (Translator) :-** It is used to convert java calls to database specific calls and database specific calls to java calls
- **Connection (Road) :-** TO communicate java with database
- **Statement (Truck) :-** To send SQL Query from java to database and to get the result
- **ResultSet (Box) :-** Output of SQL query

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